

Economics 659-600 – Behavioral Game Theory

Dr. John Van Huyck – Spring 2014

Organization: The class meets from 11:55 to 1:10 on Tuesdays and Thursdays in Allen 1004. I will have office hours from 10:00 to 12:00 on Wednesdays in Allen 3068 or by appointment. My e-mail address is john (dot) vanhuyck (at) tamu (dot) edu.

Learning Outcomes: Upon the successful completion of this course, you will be conversant with the latest behavioral game theoretic research and know when the covered game theoretic solutions make accurate predictions and give effective advice.

Requirements: Your grade will be determined on the basis of a classroom presentation worth 30 percent, a literature survey paper worth 30 percent, and a final exam worth 40 percent. As per Student Rule 7.4, exams missed due to an unexcused absence will receive a grade of zero. Excused absences will only be granted according to the process described in Student Rule 7.

Books:

1. Colin F. Camerer, *Behavioral Game Theory: Experiments in Strategic Interaction*, (Princeton University Press, 2003) ISBN # 0-691-09039-4
2. Andreu Mas-Colell, Michael D. Whinston, and Jerry R. Green, *Microeconomic Theory*, (Oxford University Press, 1995) ISBN # 0-19-507340-1
3. Douglas D. Davis and Charles A. Holt, *Experimental Economics*, (Princeton University Press, 1993) ISBN # 0-691-04317-5
4. David Levine, *Is Behavioral Economics Doomed? The Ordinary versus the Extraordinary*, (Open Book Publishers, 2012)

Plan of the Course:

January 14: Introduction; Camerer Ch. 1
January 16: Basic Game Theory; Camerer Appendix A1.1; see also Mas-Colell Ch. 7, 8, 9
January 21: Experimental Design; Camerer Appendix A1.2; see also Davis Ch. 1
January 23: Social Preferences and Experimental Control; Camerer Ch. 2
January 28: Quantal Response Equilibrium; Camerer Ch. 3; see also McKelvey and Palfrey (1995)
January 30: Bargaining and Backwards Induction; Camerer Ch. 4
February 04: Dominance-Solvable Games; Camerer Ch. 5
February 06: Cognitive Hierarchy Models;
February 11: Learning; Camerer Ch. 6
February 13: Learning continued
February 18: Coordination; Camerer Ch. 7
February 20: Coordination continued
February 25 to April 17: Topics
April 22: Exam

Topics: The topics covered in the course is subject to negotiation. The following list includes more topics than we can cover in a one semester course.

**Backward Induction:*

M. Dufwenberg, R. Sundaram, and D. J Butler, “Epiphany in the Game of 21,” *Journal of Economic*

Behavior & Organization 75, no. 2 (2010): 132–143.

Uri Gneezy, Aldo Rustichini, and Alexander Vostroknutov, “Experience and insight in the Race game,” *Journal of Economic Behavior & Organization* 75, no. 2 (August 2010): 144-155.

I. Palacios-Huerta and O. Volij, “Field Centipedes,” *American Economic Review*, 99(4), 2009, 1619-1635.

S. Levitt, J. List, and S. Sadoff, “Checkmate: Exploring Backward Induction among Chess Players,” *American Economic Review*, 101, no. 2 (2011): 975-990.

N. McKinney and J. Van Huyck, “Eureka Learning: Heuristics and Response Time in Perfect Information Games,” *Games and Economic Behavior*, 79, May 2013, 223-232.

**Communication:*

T. N Cason, R. M Sheremeta, and J. Zhang, “Communication and efficiency in competitive coordination games,” *Institute for Empirical Research in Economics* (2010).

Crawford, Vincent P. 2007. “Let’s Talk It Over: Coordination via Preplay Communication with Level-*k* Thinking.”

P. H Kriss, A. Blume, and R. A Weber, “The Dog That Did Not Bark: Pre-Play Communication with Forgone Costly Messages” (2010).

T. Ellingsen and R. Ostling, “When Does Communication Improve Coordination?,” *The American Economic Review* 100, no. 4 (2010): 1695–1724.

A. Chaudhuri and T. Paichayontvijit, “Recommended play and performance bonuses in the minimum effort coordination game,” *Experimental Economics* (2010): 1–18.

D. J Dale and J. Morgan, “Silence is Golden: Suggested Donations in Voluntary Contribution Games,” (2010).

Comparative Statics:

A. Schmutzler, “A unified approach to comparative statics puzzles in experiments,” *Games and Economic Behavior* (2010).

Competition and Coordination Success:

T. Riechmann and J. Weimann, “Competition as a Coordination Device” (2004).

T. Riechmann, “Dynamic Behavior in Minimum Effort Coordination Games” (2005).

**Conventions:*

S. Buchner, W. Guth, and L. Miller, “Individually Selecting Among Conventions – an evolutionary and experimental analysis,” *Journal of Evolutionary Economics*, (2010).

P. Espinosa, J. Kovárik, and G. Ponti, “Strategic Interaction and Conventions” (2010).

Correlated Equilibria:

John Duffy and Nick Feltovich, "Correlated Equilibria, Good and Bad: An Experimental Study," *International Economic Review* 51, no. 3 (August 2010): 701-721.

D. Fehr, F. Heinemann, and A. Llorente-Saguer, "The Power of Sunspots: An Experimental Analysis" (2011).

**Culture:*

O. Bogach and A. Leibbrandt (2011), "An Experimental Study on the Relevance and Scope of Culture as a Focal Point."

K. Hoff, P. Pandey, and B. Brooks, "Can Social Structure Shape Preferences Detrimental to Development? Theory and Evidence from Simple Games in Rural India" (2010).

D. Engelmann and H. T. Normann, "Maximum effort in the minimum-effort game," *Experimental Economics* (2010): 1-11.

Y. Chen et al., "Which Hat to Wear? Impact of Natural Identities on Coordination and Cooperation" (2010).

A. Falk, U. Fischbacher, and S. Gaechter, "Living in Two Neighborhoods—Social Interaction Effects In The Laboratory," *Economic Inquiry* (2010).

Eliciting Beliefs:

Costa-Gomes, Miguel A., and Georg Weizsäcker. 2008. "Stated Beliefs and Play in Normal-Form Games." *Review of Economic Studies*, 75(3): 729-762.

**Field Studies of Strategic Thinking:*

A. Brown, C. Camerer, and D. Lovo, "To Review or Not Review? Limited Strategic Thinking at the Movie Box Office," 2010.

R. Ostling, J. Wang, E. Chou, and C. Camerer, "Testing Game Theory in the Field: Swedish LUPI Lottery Games," *American Economic Journal: Microeconomics*, forthcoming.

Goldfarb, Avi, and Botao Yang. 2009. "Are All Managers Created Equal?" *Journal of Marketing Research*, 46(5): 612-622.

Goldfarb, Avi, and Mo Xiao. 2011. "Who thinks about the competition? Managerial Ability and Strategic Entry in US Local Telephone Markets." *American Economic Review*, 101

Focal Points:

Ken Binmore and Larry Samuelson, "The evolution of focal points," *Games and Economic Behavior* 55, no. 1 (April 2006): 21-42.

Vincent P. Crawford, Uri Gneezy, and Yuval Rottenstreich, "The Power of Focal Points Is Limited: Even Minute Payoff Asymmetry May Yield Large Coordination Failures," *American Economic Review* 98, no. 4 (August 2008): 1443-1458.

**Guessing Games:*

Costa-Gomes, Miguel A., and Vincent P. Crawford. 2006. "Cognition and Behavior in Two-Person Guessing Games: An Experimental Study." *American Economic Review*, 96(5): 1737-1768.

Costa-Gomes, Miguel A., and Vincent P. Crawford. 2011. "Studying Cognition via Information Search in Two-Person Guessing Game Experiments." In progress.

Georganas, Sotiris, Paul J. Healy, and Roberto A. Weber 2010. "On the Persistence of Strategic Sophistication."

**Immigration:*

M. Ruta and P. E Giordani, "Coordination Failures in Immigration Policy," *World* (2011).

T. Salmon and R. Weber, "Maintaining Efficiency While Integrating Entrants From Lower-Performing Environments: An Experimental Study," (2009).

Information and Learning:

Andreas Nicklisch, "Perceiving Strategic Environments: An Experimental Study of Learning Under Minimal Information," (June 2006).

V. Semeshenko et al., "Information-driven coordination: experimental results with heterogeneous individuals," *Theory and Decision* (2010): 1–24.

Leadership and Ability:

J. Gillet, E. Cartwright, and M. Van Vugt, "Leadership in a Weak-Link Game," *Studies in Economics* (2009).

J. Brandts, D. J Cooper, and E. Fatas, "Stand by Me" (2010).

Loss Avoidance:

N. Feltovich et al., "Payoff levels, loss avoidance, and equilibrium selection in games with multiple equilibria: an experimental study" (2010).

**Networks:*

B. J Ruffle, A. Weiss, and A. Etziony, "Coordination and Critical Mass in a Network Market: An Experimental Investigation," *Working Papers* (2010).

C. Alós-Ferrer and S. Weidenholzer, "Imitation and the Role of Information in Overcoming Coordination Failures," *Vienna Economics Papers* (2010).

Amy Greenwald, Eric Friedman, and Scott Shenker, "Learning in Network Contexts: Experimental

Results from Simulations” (2000).

A. Riedl, I. Rohde, and M. Strobel, “Efficient Coordination in Weakest-Ling Games,” (2011).

**Poverty Traps:*

C. M Capra et al. (2009), “The Impact of Simple Institutions in Experimental Economies with Poverty Traps,” *Economic Journal*, 119(539) 977-1009.

V. Lei and C. Noussair (2008), Equilibrium selection in an experimental macroeconomy *Southern Economic Journal*, 74(2) 448-482.

P. D Bo, A. Foster, and L. Putterman, “Institutions and Behavior: Experimental Evidence on the Effects of Democracy,” *The American Economic Review* 100, no. 5 (2010): 2205–2229.

Predicting Initial Human Behavior:

J. R Wright and K. Leyton-Brown, “Beyond equilibrium: Predicting human behavior in normal-form games,” in *Twenty-Fourth Conference on Artificial Intelligence (AAAI-10)*, 2010.

G. Devetag and S. Di Guida, “Feature-based choice and similarity in normal-form games: an experimental study,” *DISA Working Papers* (2010).

Public Goods Games:

R. C Bayer, E. Renner, and R. Sausgruber, “Confusion and reinforcement learning in experimental public goods games,” *NRN working papers* (2009).

M. Ye et al., “One Step at a Time: Does Gradualism Build Coordination?” (2011).

Ananish Chaudhuri, “Sustaining cooperation in laboratory public goods experiments: a selective survey of the literature,” *Experimental Economics* 14, no. 1 (September 2010): 47-83.

**Quantal Response Equilibrium (QRE):*

Anderson, Simon P., Jacob K. Goeree, and Charles A. Holt. 2001. “Minimum-Effort Coordination Games: Stochastic Potential and Logit Equilibrium.” *Games and Economic Behavior*, 34(2): 177-199.

Goeree, Jacob K., Charles A. Holt, and Thomas R. Palfrey. 2005. “Regular Quantal Response Equilibrium.” *Experimental Economics*, 8(4): 347-67.

Haile, Philip A., Ali Hortaçsu, and Grigory Kosenok. 2008. “On the Empirical Content of Quantal Response Equilibrium.” *American Economic Review*, 98(1): 180-200.

Yi, Kang-Oh. 2003. “A Quantal Response Equilibrium Model of Order-Statistic Games.” *Journal of Economic Behavior and Organization*, 51(3): 413-425.

QRE Learning:

Y. Breitmoser, J. H.W Tan, and D. J Zizzo, “On the Beliefs off the Path: Equilibrium Refinement due to Quantal Response and Level-k*,” *ICBBR Working Papers* (2010).

Real Effort Weak-Link Experiments:

S. Bortolotti, G. Devetag, and A. Ortmann, “Exploring the effects of real effort in a weak-link experiment,” *CEEL Working Papers* (2009).

Stefania Bortolotti, “Incentives, group pride, and real effort in the weak-link game: An experimental analysis,” Doctoral Thesis, (2010).

**Real Time Coordination Games:*

C. Deck and N. Nikiforakis, “Perfect and Imperfect Real-Time Monitoring in a Minimum-Effort Game,” *Working Papers* (2010).

R. Oprea, K. Henwood, and D. Friedman, “Separating the Hawks from the Doves: Evidence from Continuous Time Laboratory Games” (2010).

**Social Preferences:*

M. Servatka, “Does generosity generate generosity? An experimental study of reputation effects in a dictator game,” *Journal of Socio-Economics* 39, no. 1 (2010): 11–17.

B. Skyrms and K. J.S Zollman, “Evolutionary considerations in the framing of social norms,” *Politics, Philosophy & Economics* 9, no. 3 (2010): 265.

C. F Camerer and E. Fehr, “Measuring social norms and preferences using experimental games: A guide for social scientists,” *Foundations of human sociality: Economic experiments and ethnographic evidence from fifteen small-scale societies* (2004): 55–95.

A. Cabrales et al., “Social preferences and strategic uncertainty: an experiment on markets and contracts,” *The American Economic Review* 100, no. 5 (2010): 2261–2278.

Jonathan H.W. Tan, “Religion and social preferences: An experimental study,” *Economics Letters* 90, no. 1 (January 2006): 60-67.

Strategic Thinking Horse Races:

Costa-Gomes, Miguel A., Vincent P. Crawford, and Nagore Iriberri. 2009. “Comparing Models of Strategic Thinking in Van Huyck, Battalio, and Beil’s Coordination Games.” *Journal of the European Economic Association*, 7(2-3): 365-376.

Strategic Uncertainty:

O. Andersson, C. Argenton, and J. W Weibull, “Robustness to strategic uncertainty,” *Discussion Paper* (2010).

N. Feltovich and J. Swierzbinski, “The role of strategic uncertainty in games: an experimental study of cheap talk and contracts in the Nash demand game,” *European Economic Review* (2010).

**Strategy Method:*

P. Dal Bó and G. R Fréchette, “Strategy Choice In The Infinitely Repeated Prisoners Dilemma” (2010).

P. D Bo and G. R Fréchette, “The evolution of cooperation in infinitely repeated games: Experimental evidence,” *The American Economic Review* 101, no. 1 (2011): 411–429.

Jordi Brandts and Gary Charness, “The strategy versus the direct-response method: a first survey of experimental comparisons,” *Experimental Economics* (January 2011).

Team Decision Making:

Burchardi, Konrad B., and Stefan P. Penczynski. 2010. “Out Of Your Mind: Eliciting Individual Reasoning in One Shot Games.”

Cooper, David J., and John H. Kagel. 2005. “Are Two Heads Better Than One? Team versus Individual Play in Signaling Games.” *American Economic Review*, 95(3): 477-508.

**The Winner's Curse:*

Charness, Gary and Dan Levin. 2009. “The Origin of the Winner’s Curse: A Laboratory Study.” *American Economic Journal: Microeconomics*, 1(1): 207-236.

Ivanov, Asen, Dan Levin, and Muriel Niederle. 2010. “Can Relaxation of Beliefs Rationalize the Winner’s Curse?: An Experimental Study.” *Econometrica*, 79(4): 1435-1452.

Costa-Gomes, Miguel A., and Makoto Shimoji. 2010. “A Comment on ‘Can Relaxation of Beliefs Rationalize the Winner’s Curse? An Experimental Study’”. In progress.

Trust:

James C. Cox, Maroš Servátka, and Radovan Vadovič, “Saliency of outside options in the lost wallet game,” *Experimental Economics* 13, no. 1 (October 2009): 66-74.

Survey Papers:

G. Charness and P. Kuhn, “Lab Labor: What can labor economists learn from the lab?”, forthcoming in O. Ashenfelter and D. Card eds., *The Handbook of Labor Economics Vol. 4*.

D. Cooper and J. Kagel, “Other Regarding Preferences: A Selective Survey of Experimental Results,” (2013) forthcoming in J. Kagel and A. Roth (eds.) *The Handbook of Experimental Economics, Vol. 2*.

V.P. Crawford, “Theory and Experiment in the Analysis of Strategic Interaction,” Chapter 7 in David Kreps and Ken Wallis (eds.) *Advances in Economics and Econometrics: Theory and Applications, Seventh World Congress*, Vol. I, Cambridge 1997; reprinted in Colin Camerer, George Loewenstein, and Matthew Rabin, (eds.) *Readings in Behavioral Economics*, Princeton and Russell Sage Foundation, February 2004.

V.P. Crawford, M. A Costa-Gomes, and N. Iriberri, "Strategic thinking," (2010).

J. Duffy, "Macroeconomics: A Survey of Laboratory Research," forthcoming in J. Kagel and A. Roth (eds.) *The Handbook of Experimental Economics*, Vol 2.

I. Erev and E. Haruvy, "Learning and the Economics of Small Decisions," forthcoming in J. Kagel and A. Roth eds. *The Handbook of Experimental Economics*, Vol 2.

D. Fudenberg and D. Levine, "Learning and Equilibrium," *The Annual Review of Economics*, (2009) 385-419.

G.W. Harrison and J.A. List, "Field Experiments," *Journal of Economic Literature*, 42(4), 2004, 1009-1055.

J. Kagel and A. Roth, *The Handbook of Experimental Economics*, Princeton 1995.

Required Statements:

"Aggie Honor Code: 'An Aggie does not lie, cheat, or steal or tolerate those who do.' Upon accepting admission to Texas A&M University, a student immediately assumes a commitment to uphold the Honor Code, to accept responsibility for learning, and to follow the philosophy and rules of the Honor System. Students will be required to state their commitment on examinations, research papers, and other academic work. Ignorance of the rules does not exclude any member of the TAMU community from the requirements or the processes of the Honor System. For additional information please visit: <http://student-rules.tamu.edu>."

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